MILITARY SPECIFICATION

IGNITER, PRIMARY

This specification is approved for use by the Naval Sea Systems Command (OS), Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Igniter, Primary, for use in aircraft ejection seat catapults.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATION

FEDERAL

TT-I-1795	Ink, Marking Stencil, Opaque (Porous and Non-Porous Surfaces)

MILITARY

MIL-P-116	Preservation-Packing, methods of
MIL-C-14884	Catapult, Aircraft Ejection Seat M37 Assembly
MIL-C-46395	Catapult, Aircraft Ejection Seat, M9 Assembly
MIL-C-60539	Coatings, Anodic, Conventional, For Aluminum and Aluminum Alloys.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Division (611), Indian Head, Maryland 20640 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking For Shipment and Storage
MIL-STD-453	Inspection, Radiographic
MIL-STD-810	Environmental Test Methods
MIL-STD-1167	Ammunition Data Card

PUBLICATIONS

NAVAL SEA SYSTEMS COMMAND (CODE IDENT 10001)

OS 9765	Boron-Potassium Nitrate

DRAWINGS

NAVAL ORDNANCE STATION, INDIAN HEAD, MD (CODE IDENT 14083)

8856866	Carton, Packing, Ammunition, For Igniter, Primary Assembly, For M9 and M37 Rocket Catapult
8856867	Box, Packing, Ammunition, For Igniter, Primary Assembly, For M9 and M37 Rocket Catapult
10524201	Igniter, Primary Assembly

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ANSI B46.1	Surface Texture

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018)

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Code of Federal Regulations

49 CFR 100-199 Transportation

(The Code of Federal Regulations is available from the Superintendent of Documents, Government Printing Office, Washington, DC 20402. Orders should cite "49 CFR 100-199 (latest revision)".)

3. REQUIREMENTS

3.1 First article. Unless otherwise specified in the contract (see 6.2), first article samples of the igniters are required and shall be manufactured using the methods and procedures proposed for production. The samples shall be tested as specified in Section 4 for the purposes of determining, prior to starting production, that the contractor's production methods are capable of producing igniters that comply with the requirements specified on the applicable drawings and referenced specifications.

3.2 Materials. All materials, parts and assemblies shall be in accordance with 10524201, this specification, and the documents referenced herein. When drawings or specifications permit the use of equivalent items or materials, authorization to use a specific alternate shall be obtained from the Government procuring activity in writing.

3.3 Product characteristics and performance requirements.

3.3.1 Visual examination. The igniter shall meet the requirements of drawing 10524201 when examined in accordance with 4.5.1.

3.3.2 Radiographic inspection. Radiographic inspection of the igniter shall confirm the presence and arrangement of components to be in compliance with applicable drawings and specifications when examined in accordance with 4.5.2. Igniters scheduled for vibration testing shall be radiographically inspected before and after vibration and shall meet these requirements both times.

3.3.3 Vibration. The igniter shall withstand the vibration test of 4.5.3 without any disintegration of the primer or igniter charge, or the disarrangement of any component. (This requirement applies only to the first article sample of igniters.)

3.3.4 Protective finish. The protective finish shall be in accordance with MIL-C-60539, Type II, Level 1, when examined in accordance with 4.5.4.

3.3.5 Sealant. Sealant application shall meet the drawing requirements when examined in accordance with 4.5.5.

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3.3.6 Pressure-time requirements. The igniters shall meet the pressure-time requirements of TABLE I when assembled into inert loaded rocket ejection seat catapults M9 or M37 and tested in accordance with 4.5.6.

TABLE I. Pressure-time requirement.

Parameter	Firing temperatures					
	-65 to -70°F		70 ± 5°F		165 ± 5°F	
	MAX	MIN	MAX	MIN	MAX	MIN
Maximum pressure (psig)	471	315	498	290	488	290
Time to maximum pressure (ms)	53	19	43	21	41	20
Pressure rise rate (psig/ms)	69	0	55	2	64	0

3.3.7 Additional requirements. The igniters shall function when assembled rocket ejection seat catapults M9 or M37 are undergoing the lot acceptance ballistic tests of MIL-C-46395 or MIL-C-14884, respectively (see 4.5.7.)

3.3.8 Marking. Each igniter shall be clearly and permanently identified by stamping or stenciling with the igniter designation, identifying number, serial number, and lot number. No other markings shall appear thereon except those specified herein or on the applicable drawing.

3.4 Workmanship. All parts shall be free of chips, dirt, grease, rust and all other foreign materials. There shall be no burrs or other imperfections of manufacture which would affect the assembly, functioning or handling of the product. The cleaning methods used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)

b. Quality conformance inspection (see 4.4)

4.3 First article sample. Unless otherwise specified, at the beginning of regular production a sample shall be submitted in accordance with contract requirements and shall consist of 26 assembled igniters, 10 sets of unassembled parts (including explosive components) for visual and dimensional examination and 23 sets of unassembled components without explosive components for protective finish tests. The sample shall be manufactured using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

4.3.1 Examination and test. After inspection and provisional acceptance at source, the sample shall be inspected at a Government laboratory or such other facility specified in the contract for conformance to all the requirements of the drawings and specifications. The sample shall be subjected to the examinations and tests of 4.5.

4.3.2 First article sample failure. Failure of the sample to meet the requirements of the drawings and specifications shall result in sample disapproval.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the examinations and tests of 4.5. Failure of the sample to meet the requirements of the drawings and specifications shall be cause for rejection of the lot.

4.4.1 Test reports. When specified in the contract (see 6.2), the contractor shall furnish test reports to the procuring activity. Test reports shall include the following information:

a. Protective finish (see 4.5.4)
b. Radiographic examination (see 4.5.2)
c. Sealant application (see 4.5.5)
d. Ignition material (see 4.6)
e. Heat of explosion (see 4.6)
f. Differential thermal analyses (see 4.6)

4.4.2 Lot formation. For the purpose of identification and shipment, the shipping lots of loaded igniter assemblies shall consist of 1200 assemblies, maximum, exclusive of test samples, provided the homogeneity of the inspection lots has been retained as follows:

a. Components produced by a homogeneous process
b. Propellant from not more than one lot
c. Igniter from not more than one lot
d. Primers from not more than one lot

A smaller lot may be shipped upon approval of the contracting officer.

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4.4.2.1 Sampling for examination. Samples for examination shall follow MIL-STD-105, general inspection level II.

4.4.2.2 Sampling for ballistic (pressure-time) performance. A sample of 21 igniters shall be submitted to the Naval Ordnance Station, Indian Head, MD 20640. The igniters shall not be preselected during production.

4.5 Examination and tests.

4.5.1 Examination. One hundred percent examination shall be performed for all critical defects. Examination for major or minor defects shall be performed on a defect basis in accordance with the classification of defects, using the sampling plan of MIL-STD-105, general inspection level II. The total number of allowable defects per class for either the major or minor class shall not exceed an AQL level of 1.0% and 1.5% respectively. Failure to meet either requirements shall be cause for rejection of the lot. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burrs and foreign matter. The roughness comparison specimens prescribed by ANSI-B46.1 shall be used as a basis of comparison for surface roughness determination.

4.5.1.1 Primer, 72M, 7645332, covering a detail of Drawing 10524201

Defect	Method of Inspection
CRITICAL	
1. Mixed primers	Visual
2. Primer pellet dislodged	Visual
3. Primer foil discolored	Visual
4. Metal defective	Visual
MAJOR	
None defined	
MINOR	
None defined	

4.5.1.2 Igniter, Primary, Drawing 10524201

CRITICAL	
1. Mixed igniters	Visual
2. Metal defective	Visual
3. Marking missing, incorrect or illegible	Visual

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Igniter, Primary, Drawing 10524201 (continued)

Defect		Method of Inspection
MAJOR		
101.	Profile incorrect	Gage
102.	Primer above surface "S"	Tactile
103.	Pocket crimp missing or incorrect	Visual
104.	Foreign matter	Visual
105.	Burr	Visual
MINOR		
201.	Diameter of recess, min.	Gage
202.	Dent	Visual
203.	Draw scratch	Visual
204.	Fold, wrinkle, buckle or bulge	Visual
205.	Varnish inadequate or missing	Visual
206.	Surface finish improper	Visual

4.5.1.3 Carton, Packing, Drawing 8856866

Defect		Method of Inspection
CRITICAL		
	None defined	
MAJOR		
	None defined	
MINOR		
201.	Contents loose	Visual
202.	Quantity of contents incorrect	Visual
203.	Separation or filler missing	Visual
204.	Heat sealer incomplete, loose or distorted	Visual
205.	Cut or damaged through all impregnated layers	Visual
206.	Marking missing, incorrect or illegible	Visual

4.5.1.4 Box, Packing, Drawing 8856867

Defect		Method of Inspection
CRITICAL		
	None defined	
MAJOR		
	None defined	

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Box, Packing, Drawing 8856867 (continued)

Defect	Method of Inspection
MINOR	
201. Contents loose	Visual
202. Hardware missing, loose, broken or improperly engaged	Visual
203. Strapping missing, loose or improperly assembled	Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of cartons incorrect	Visual
208. Car seal missing, unsealed or improperly positioned	Visual

4.5.2 Radiographic inspection. All igniters shall be radiographically inspected in accordance with MIL-STD-453, radiographic quality level 1 at ambient temperature. Any defect shall be cause for rejection of the igniter. Radiographs shall be capable of determining proper seating of primer, proper installation and crimping of igniter disc, and proper loading of ignition pellets. When specified in the contract (see 6.2), one set of radiographs taken of each igniter in the lot shall be forwarded to the Naval Ordnance Station, Indian Head, MD 20640. Acceptance of the igniter lot shall be subject to approval of the radiographs by the Naval Ordnance Station. The assembled igniter shall be x-rayed with the long axis of the igniter perpendicular to the rays of the x-ray machine. Failure of any igniter to conform to the drawing requirements shall be cause for rejection of the igniter.

4.5.3 Vibration. Each of 21 igniter assemblies of the initial production sample shall be subjected to vibration. The igniters shall then be x-rayed. Failure of any igniter to comply with the requirements of 3.3.3 shall be cause for rejection of the lot. Any change in the process of a lot of primers in subsequent production shall be reason to select an additional sample of 21 igniter assemblies for vibration tests. The igniter assembly shall be subjected to random vibration in accordance with MIL-STD-810, Method 514.2, "Random Vibration Test", for 6 hours, 2 hours along each axis. The samples shall be temperature conditioned for the test; seven igniters each at -65 to -70°F, 70 ± 5°F and 165 ± 5°F. The input acceleration shall be in accordance with MIL-STD-810, figure 514.2-2A, curve A, with a "Wo" factor of .1G^2/Hz. If the systems must be removed from the environmental conditioning chamber or if the temperature falls outside the specified temperature limits for a period of more than 5 minutes, the testing shall be stopped and the systems reconditioned. A reconditioning time of 2 minutes for every minute the systems are outside the specified temperature envelope shall be required. A continuous temperature history of the systems shall be kept for all conditioning and

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testing time. Unless otherwise noted, the systems shall be mounted on the test fixture, preloaded in a manner that will stimulate actual catapult installation and excited through the attachment fittings.

4.5.4 Protective finish. Twenty-three units of each component shall be selected for protective finish tests. Test procedures for protective finish shall be in accordance with MIL-C-60539. The igniter case shall be sectioned longitudinally to better expose the significant surfaces to the salt spray. Failure of any component to comply with the requirements of 3.3.4 shall be cause for rejection of the lot.

4.5.5 Sealant application. Five igniter assemblies from each initial production sample and each production lot shall be selected for this test. Failure of any igniter to comply with the drawing requirements shall be cause for rejection of the lot of igniter assemblies. The igniter assembly shall be disassembled in such a manner that determination can be made that the sealant has been applied for the full distance of the adjoining surfaces of the disc and the case sidewall and that the propellant cavity is free of sealant.

4.5.6 Pressure-time test. A sample of 21 igniters, after being radiographically examined, vibrated, and radiographically examined again, shall be assembled into inert loaded rocket ejection seat catapults M9 or M37. The assembled catapults shall be temperature conditioned for 8 hours at the temperatures shown in TABLE I (7 at each temperature) and then test fired within 10 minutes after removal from the conditioning chambers. Failure to meet any parameter of TABLE I shall be cause for rejection of the lot. The loading and firing of these units shall be performed by and at the Naval Ordnance Station, Indian Head.

4.5.7 Additional test requirement. Igniters used in the lot acceptance ballistic testing of rocket ejection seat catapults M9 or M37 in accordance with MIL-C-46395 or MIL-C-14884, respectively, shall not fail to function. Failure of an igniter to function shall be cause for rejection of the lot.

4.6 Ignition material. At the time the igniters are submitted for testing and when specified in the contract (see 6.2), the supplier shall supply the following ignition material information accompanied by a certificate which attests that the information is correct and applicable to the product submitted:

a. A statement that the ignition material complies with all requirements and quality assurance provisions of OS 9765

b. Certificates of analyses including heat-of-explosion test and differential thermal analysis results.

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5. PACKAGING

5.1 Preservation and packaging. Preservation and packaging shall be level A.

5.1.1 Packaging. The igniters shall be packaged in containers subject to the approval of Naval Ordnance Station, Indian Head, MD. The package shall be cushioned adequately to protect the igniters against handling and shipping shocks and vibrations, and shall comply with the applicable requirements of MIL-P-116 and in accordance with 49 CFR 171-190.

5.2 Packing. Packing shall be level A.

5.2.1 Shipping containers. The igniters shall be placed into support cartons as specified in drawing 8856866 and packed in boxes conforming to drawing 8856867, and in accordance with 49 CFR 171-190. Cushioning shall be applied as required to obtain a tight pack.

5.3 Marking. In addition to any special marking required by the contract, unit packages and single containers shall be marked in accordance with the requirements of MIL-STD-129.

5.3.1 Special marking. Each contained igniter (see 5.1.1) shall be labeled with the following warning in 1/8-inch minimum high letters using black stencil ink conforming to TT-I-1795 WARNING: This material is ignited from sparks, flame, or handling.

5.4 Data cards. Data cards, DD Form 1650, shall be furnished and filled out in accordance with MIL-STD-1167 by the supplier for each production lot of ignition material. Two data cards shall be placed in each shipping container. Additional data cards shall be provided as directed by the Government representative or the Government procuring activity.

6. NOTES

6.1 Intended use. The igniter is intended for use as a primary ignition charge for rocket catapults, M9 and M37.

6.2 Ordering data. Procurement documents shall specify, but not be limited to, the following:

6.2.1 Procurement requirements.

a. Title, number, and date of this specification
b. First article sample if other than specified (see 4.3)
c. Test facility responsible for first article testing (see 4.1 and 4.3)
d. Test conditions if other than specified
e. If test reports are required (see 4.4.1)
f. If radiographs of the igniters are to be submitted (see 4.5.2)
g. Test facility responsible for ballistics testing if other than the Naval Ordnance Station, Indian Head (see 4.4.2.2 and 4.5.6)
h. Safety precautions (see 6.3)

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6.2.2 Contract data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Armed Services Procurement Regulations, the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with Approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of ASPR-7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
4.4.1	Test Reports	-
4.5.2	Radiographs of igniters	-
5.4	Ammunition Data Card	DI-E-2001

(Copies of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.3 Safety precautions. The safety precaution requirements of the "Contractors' Safety Manual for Ammunition, Explosives and Related Dangerous Material" (DOD 4145.26M) are applicable and should be specified in the contract or order as required by the Armed Services Procurement Regulations (ASPR) 1-323.

NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore" (OP 5) should be made applicable.

6.4 Warning. The ignition material, when dry, is a dangerous, highly sensitive explosive. All tests should be accomplished in a room set aside for work on primary explosives. Extreme precautions should be taken in all stages of work, and special care should be taken to avoid accumulation of static charge. Conductive containers should be used when handling the dry ignition material. No quantity of ignition material larger than the minimum quantity necessary should be taken at any one time.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Navy - OS

Preparing Activity:
Navy - OS

Project No. 1377-N670

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-I-60676A(OS) IGNITER, PRIMARY

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED S/N 0102-014-1802

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